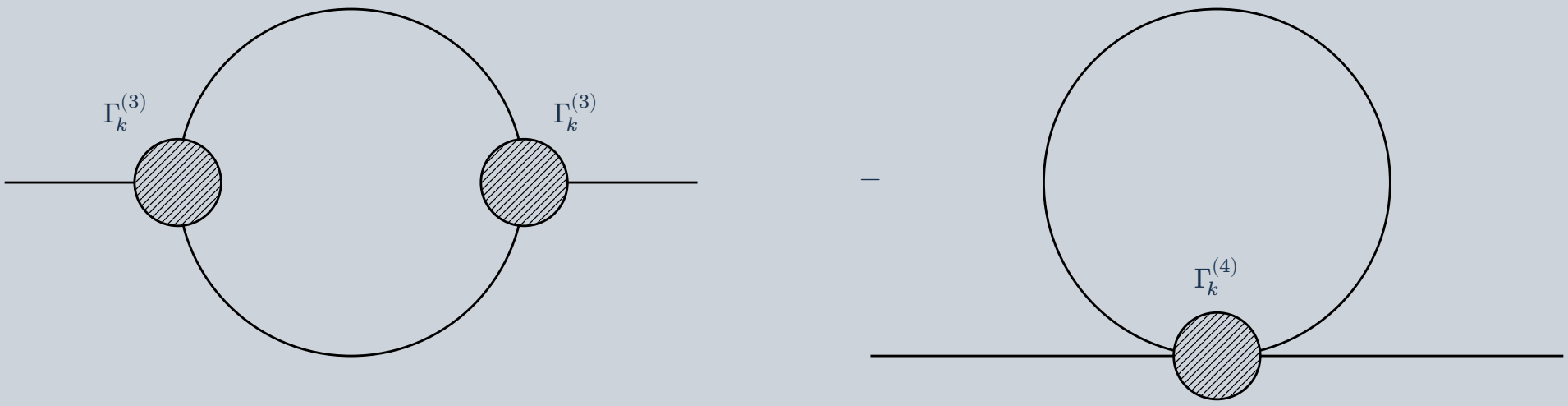
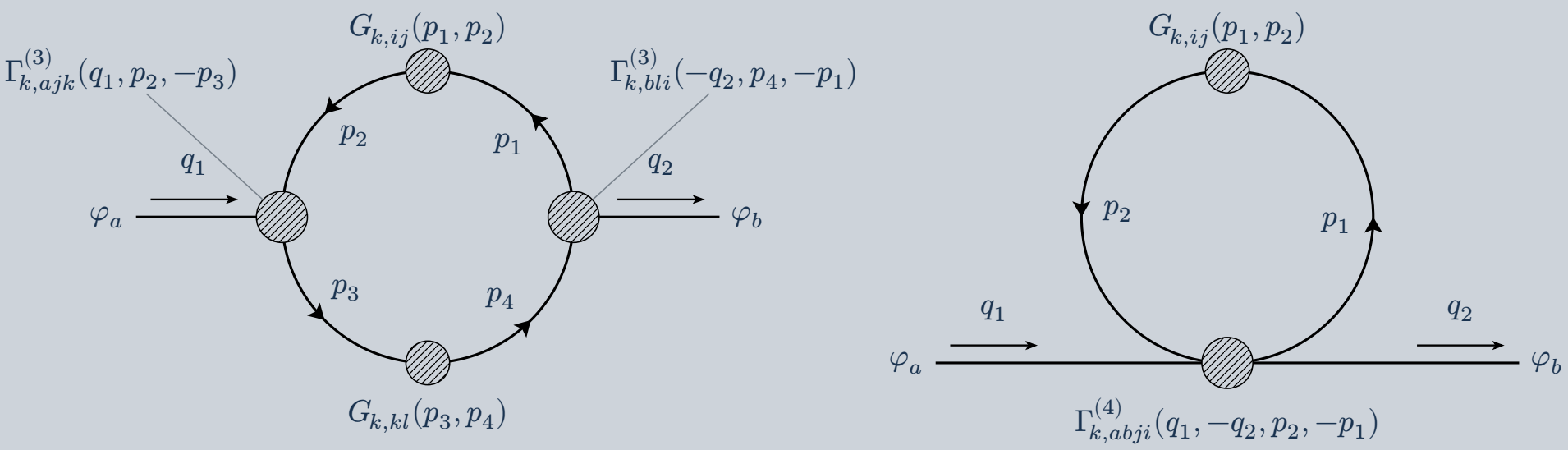
Compare the loop structures before and after inserting a scale-dependent regulator. The compact view introduces the topology; the detailed views show momentum and field indices.

1 Recognize the two topologies



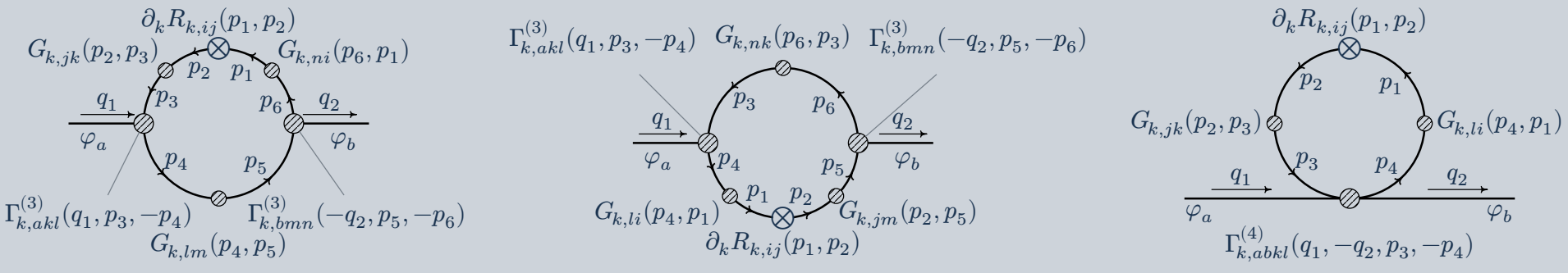
A loop with two three-point vertices and a tadpole with one four-point vertex. Signs and symmetry factors belong to the defining equation and conventions.

2 Resolve the internal labels



The same building blocks with explicit external legs and loop momenta. Without a regulator, integration is not restricted by the running cutoff.

3 Insert the changing cutoff



Each circled cross is $\partial_k R_k$. Different insertion positions contribute to the scale derivative of the two-point function; these sketches show the terms, not their prefactors.

The cross makes a flow diagram scale selective. With a suitable regulator, low-momentum modes are suppressed and the derivative weights the modes being integrated out near scale k. G_k is a full propagator; $\Gamma_k^{(n)}$ couples n field legs. Hatched discs mark these dressed propagators and vertices.